Computer Organization and Architecture
(Code: CS256)

Lecture-29: External Memory Design II

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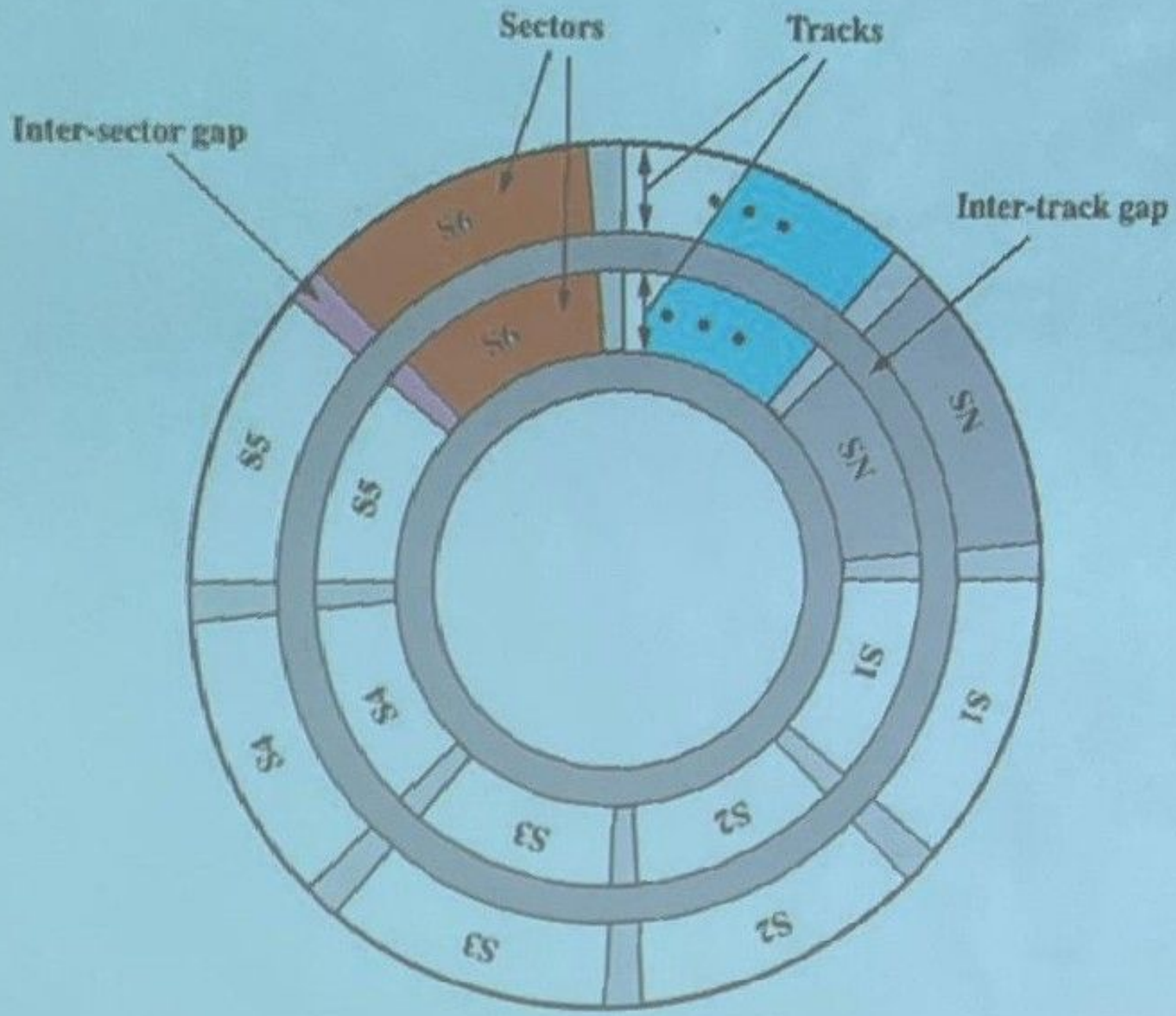
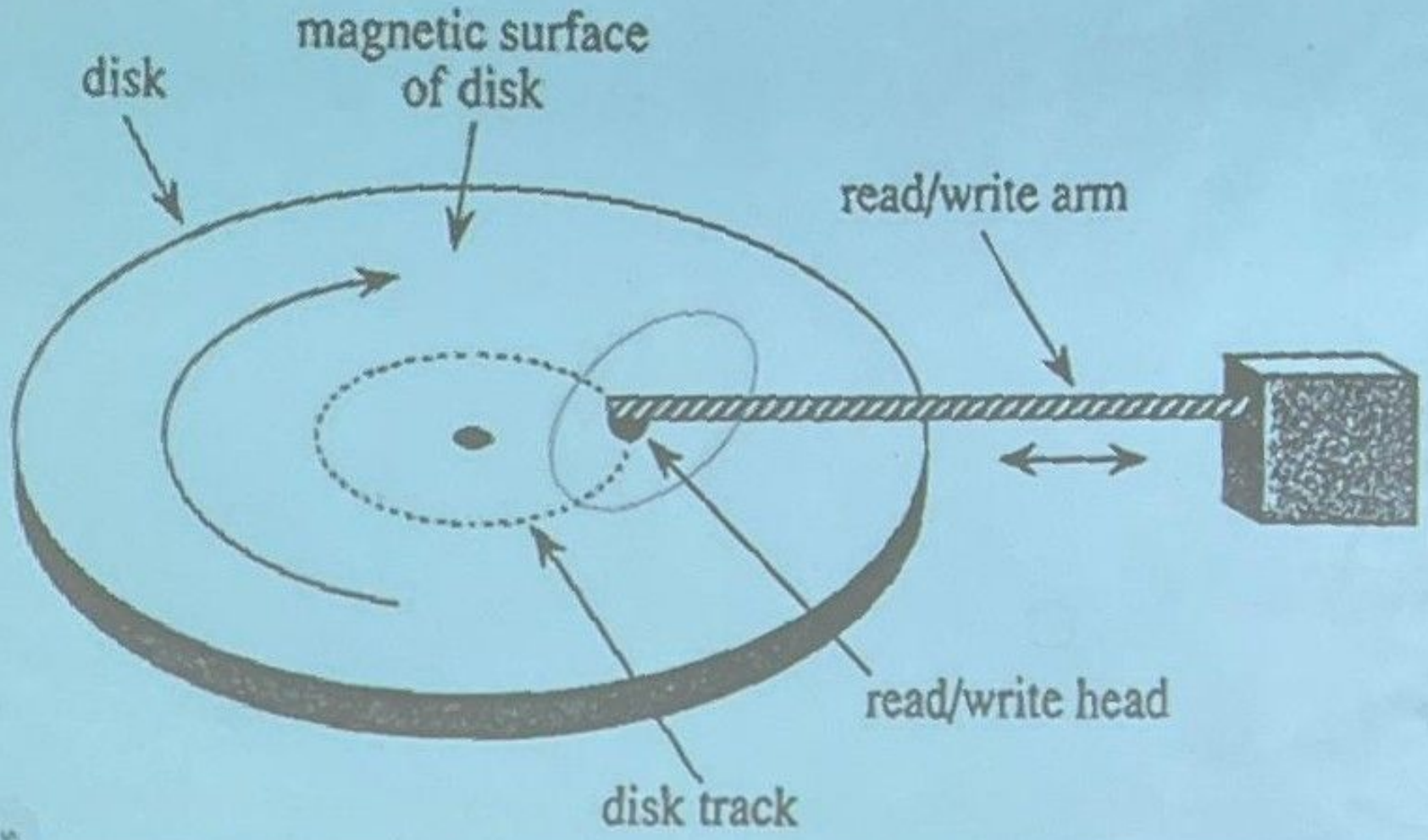
Disk Performance

Data Organization and Formatting



Read and Write Mechanisms

- Recording (write) and retrieval (read) via conductive coil called a head.
- May be single read/write head or separate ones.
- During read/write, head is stationary, platter rotates

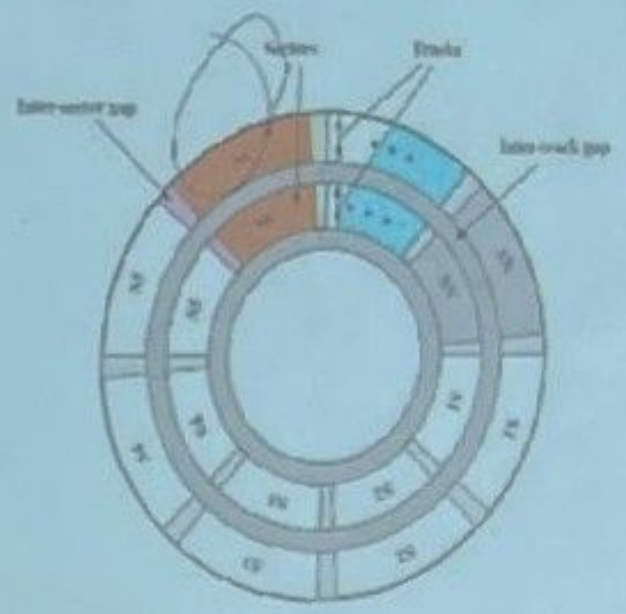


Typical Hard Disk Drive Parameters

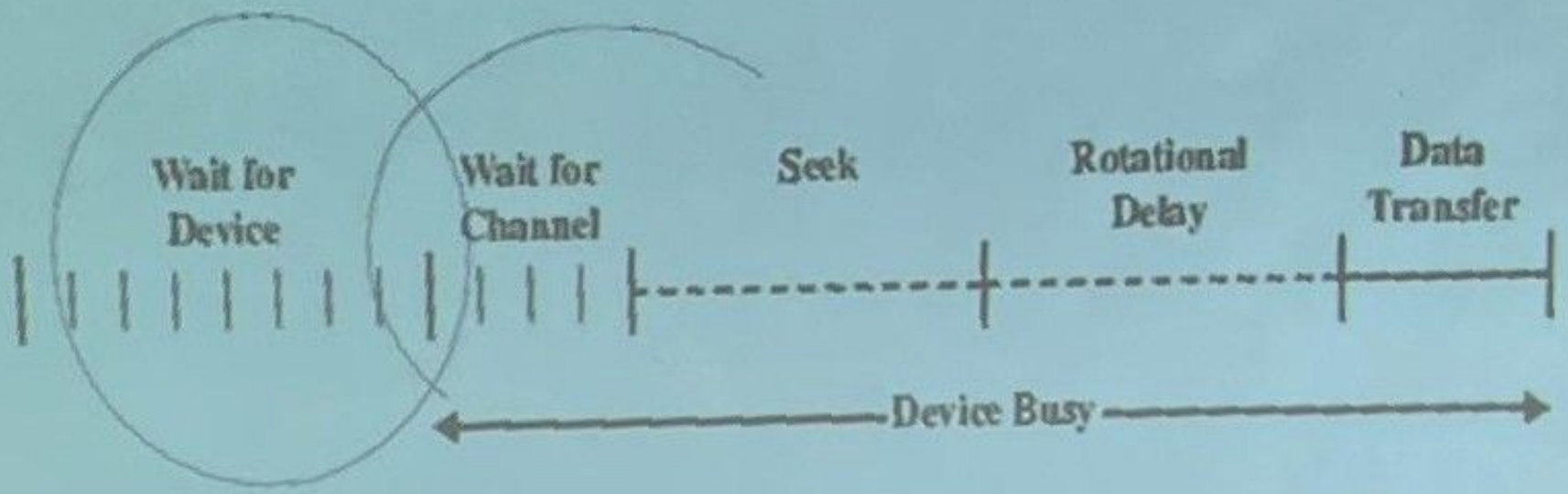
Characteristics	Seagate Barracuda 180	Seagate Cheetah X15-36LP	Seagate Barracuda 36ES	Toshiba HDD1242	Hitachi Microdrive
Application	High-capacity server	High-performance server	Entry-level desktop	Portable	Handheld devices
Capacity	181.6 GB	36.7 GB	18.4 GB	5 GB	4 GB
Minimum track-to-track seek time	0.8 ms	0.3 ms	1.0 ms	—	1.0 ms
Average seek time	7.4 ms	3.6 ms	9.5 ms	15 ms	12 ms
Spindle speed	7200 rpm	15K rpm	7200	4200 rpm	3600 rpm
Average rotational delay	4.17 ms	2 ms	4.17 ms	7.14 ms	8.33 ms
Maximum transfer rate	160 MB/s	522 to 709 MB/s	25 MB/s	66 MB/s	7.2 MB/s
Bytes per sector	512	512	512	512	512
Sectors per track	793	485	600	63	—
Tracks per cylinder (number of platter surfaces)	24	8	2	2	2
Cylinders (number of tracks on one side of platter)	24,247	18,479	29,851	10,350	—

Speed Parameters

- **Seek time**
  - Moving head to correct track
  - Typical average seek time is under 10 ms
- **(Rotational) latency**
  - Waiting for data to rotate under head
  - 15,000 rpm (rotation per minute) = 250 r/sec
  - 1/250 = 0.004 sec → There is one revolution per 4ms.
  - Thus, average rotation delay (1/2r) : about 2ms
- **Transfer rate**
  - Time required for data transfer
- **Additional delays**
  - Waiting for I/O channels
  - Rotation Positional Sensing (RPS)



Timing of Disk I/O Transfer



Speed Parameters

- The **transfer time** to or from the disk depends on the rotation speed of the disk in the following fashion:

$$T = (b/N)*(1/r)$$
$$= b/rN$$

- b : # of bytes to be transferred
- N : # of bytes on a track
- r : rotation speed, in revolutions per second

Speed Parameters

- **Total average access time**

$$Ta = Ts + 1/(2r) + b/(rN)$$

  - Ts : average seek time
  - 1/(2r) : average rotation delay
  - b/(rN) : transfer time

Disk Performance

- Disk parameters
  - average seek time : 4ms
  - rotation speed : 7,200 rpm (average rotation delay = 4ms)
  - 512 B/sector
  - 500 sector/track

2,500 sectors → 5 adjacent tracks

•best case:

For 1st track: 1/(2r)=4ms → 1/r : 1 track transfer time:8ms

For 4 succeeding tracks : no seek time needed. 4 tracks * (avg rotation delay:4 + track transfer time) = 4 * (4ms + 8ms)

Disk Performance

- Read 2,500 sectors
  - Best case
  - In the case that the file is stored as compactly as possible on the disk. That is, the file occupies all of the sectors on 5 adjacent tracks (5 tracks X 500 sectors/track = 2,500 sectors)
  - The time to read the first track :
    - 4ms(average seek) + 4ms(average rotational delay) + 8ms(first track transfer)
  - Suppose that the remaining tracks(4 tracks) can now be read with essentially no seek time:
    - + 4x(4+8)
  - Total time:
    - 4 + 4 + 8 + 4 x(4 + 8) = 64 ms

Disk Performance

- Read 2,500 sectors
  - Best case
    - 4ms(seek) + 4ms(rotation) + 8ms(first track transfer) + 4x(4+8) = 64 ms
  - Worst case
  - Now let's calculate the time required to read the same data using random access than sequential access; that is, accesses to the sectors are distributed randomly over the disk:
  - For each sector, we have
    - 4ms(seek) + 4ms(rotation) + 0.016 ms(1 sector transfer)
  - Total time for 2,500 sectors,
    - 2,500 x 8.016 ms = 20.04 sec

RAID

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RAID

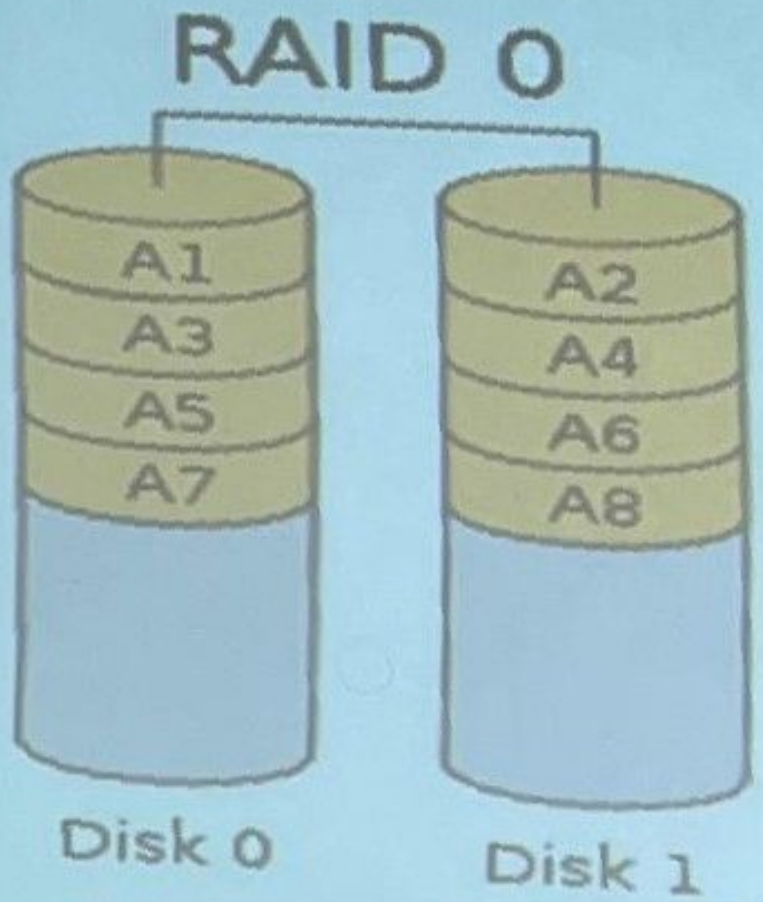
- RAID stands for
  - *Redundant Array of Independent Disks*
  - *Redundant Array of Inexpensive Disks*
- *RAID is a data storage virtualization technology that combines multiple physical disk drive components into one or more logical units for the purposes of data redundancy, performance improvement, or both.*
- First used in 1988 by Patterson, Gibson, and Katz from Berkeley
- Data is distributed across the drives in one of several ways, referred to as RAID levels.
- Levels are Not in a hierarchy.
- Depending on the required level of redundancy and performance.
  - 6-7 levels in common use

RAID-Levels

- The different schemes, or data distribution layouts, are named by the word "RAID" followed by a number, for example RAID 0 or RAID 1.
- *Each scheme, or RAID level, provides a different balance among the key goals: reliability, availability, performance, and capacity.*
- RAID levels greater than RAID 0 provide protection against unrecoverable sector read errors, as well as against failures of whole physical drives.
- Set of physical disks viewed as single logical drive by O/S
- Data distributed across physical drives
- Can use redundant capacity to store parity information

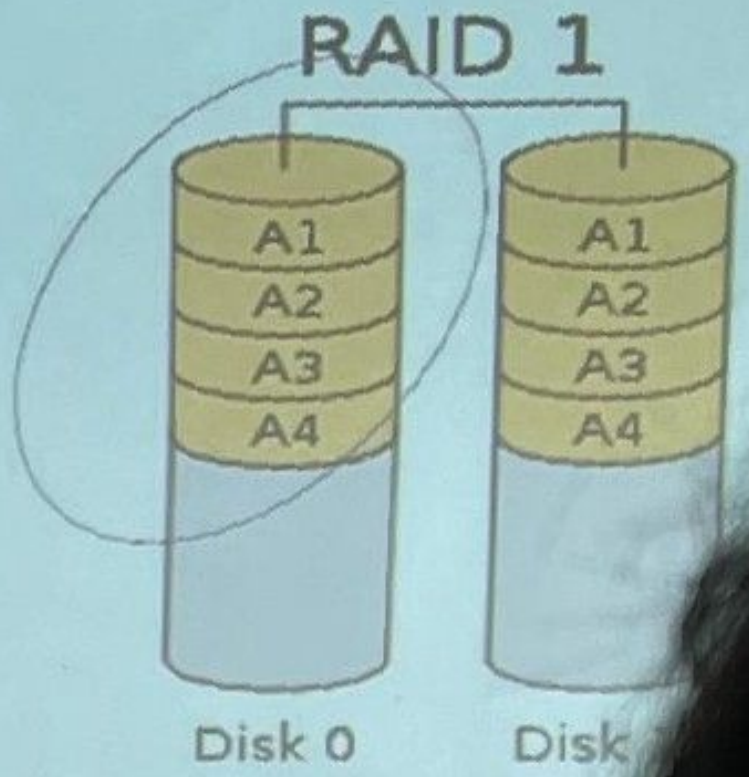
RAID-0

- This configuration has striping *but no redundancy of data.*
- It offers the **best performance**, but it does *not provide fault tolerance.*



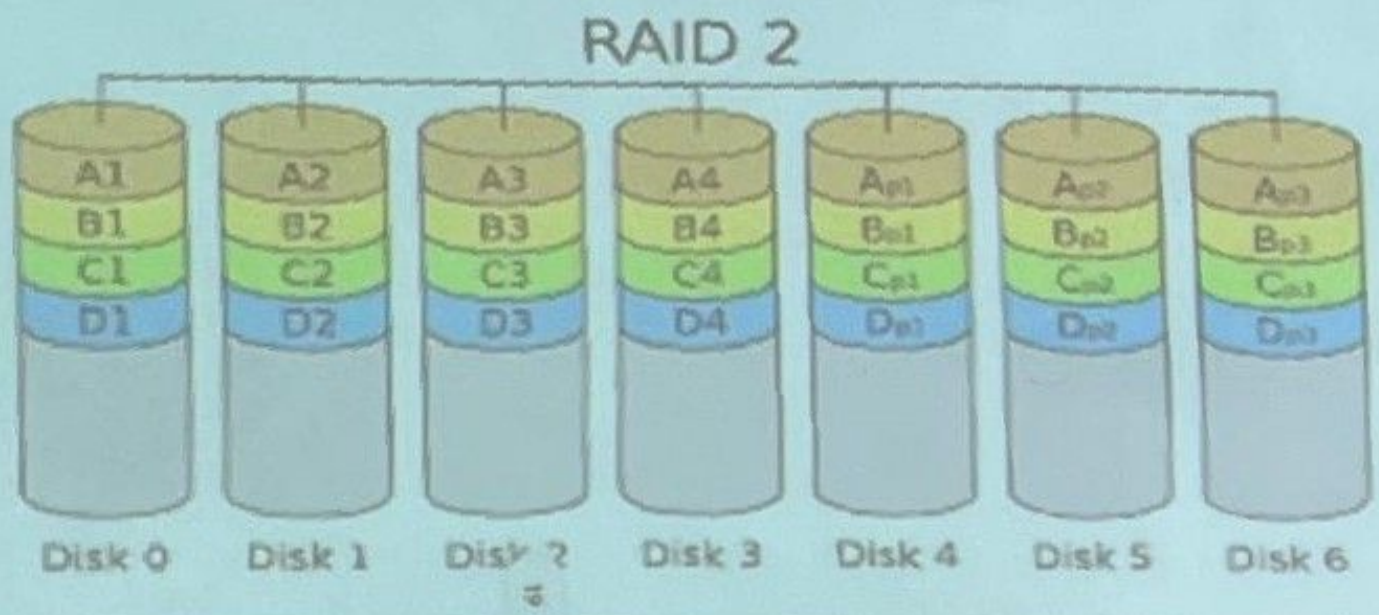
RAID-1

- *Also known as disk mirroring.*
- This configuration consists of at least two drives that duplicate the storage of data.
- There is no striping.
- Read performance is improved, since either disk can be read at the same time.
- Write performance is the same as for single disk storage.



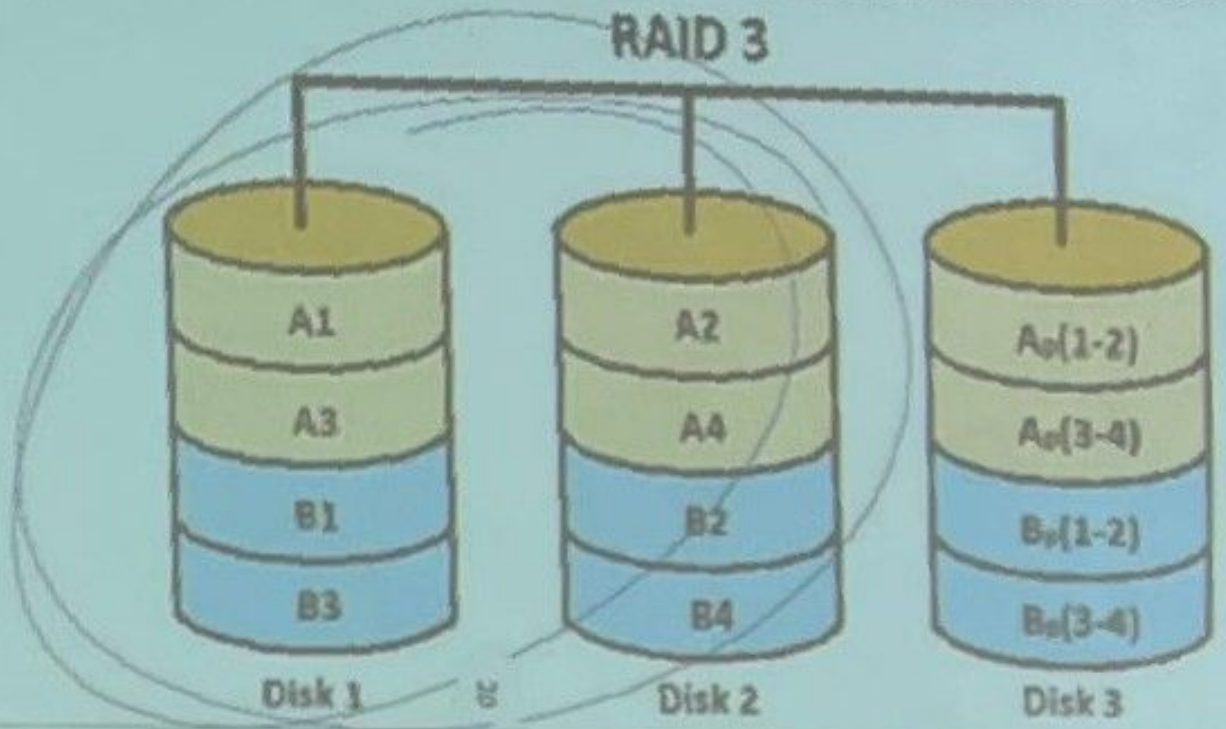
RAID-2

- This configuration uses striping across disks, with some disks storing error checking and correcting (ECC) information.
- RAID 2 also uses a dedicated *Hamming code* parity, a linear form of ECC.
- RAID 2 has no advantage over RAID 3 and is no longer used.



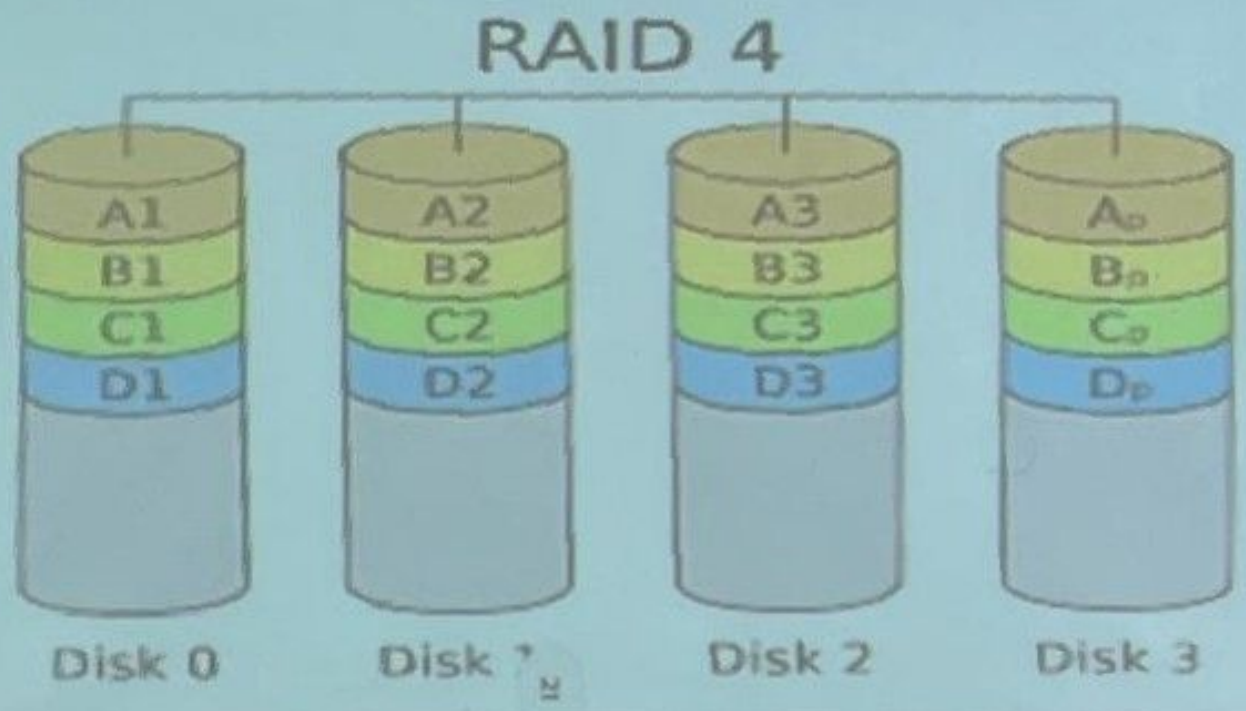
RAID-3

- This technique uses striping and dedicates one drive to storing **parity** information.
- The embedded ECC information is used to detect errors.
- Data recovery is accomplished by calculating the exclusive information recorded on the other drives.
- Because an I/O operation addresses all the drives at the same time, RAID 3 cannot overlap I/O.
- ***For this reason, RAID 3 is best for single-user systems with long record applications.***



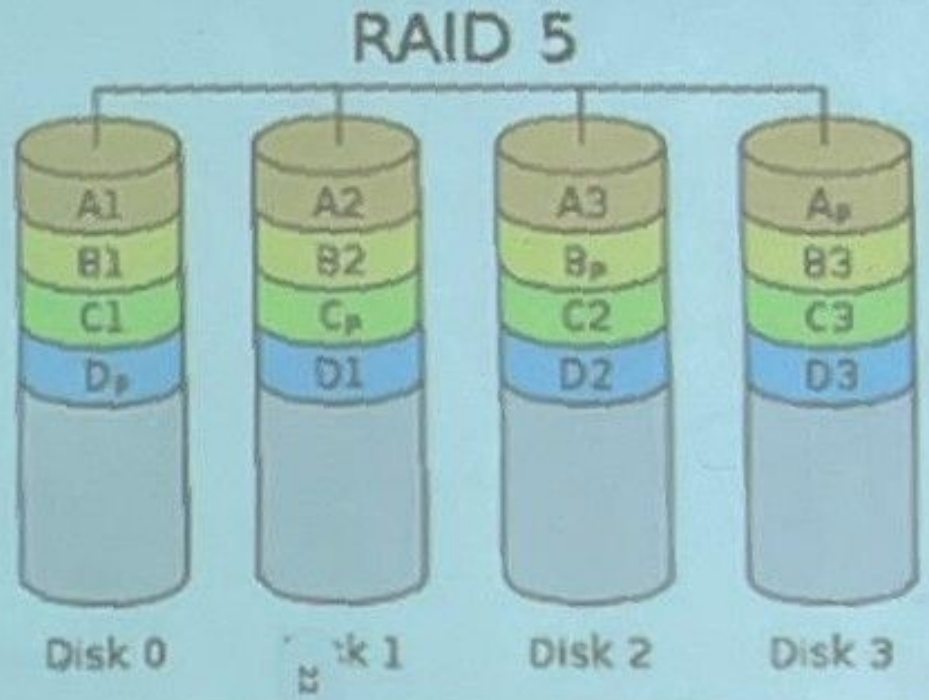
RAID-4

- This level uses large stripes, which means a user can read records from any single drive.
- Overlapped I/O can then be used for read operations.
- Because all write operations are required to update the parity drive, no I/O overlapping is possible.



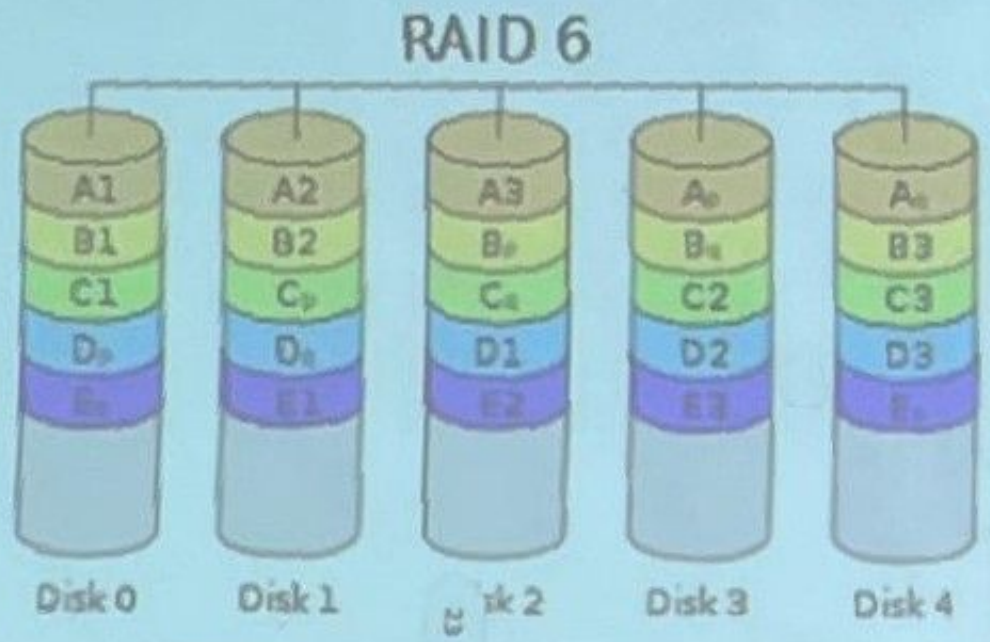
RAID-5

- This level is based on parity block-level striping.
- The parity information is striped across each drive, enabling the array to function, even if one drive were to fail.
- The array's architecture enables read and write operations to span multiple drives.
- This results in performance better than that of a single drive, but not as high as a RAID 0 array.
- *RAID 5 requires* ***at least three disks****, but it is often recommended to use at least five disks for performance reasons.*



RAID-6

- This technique is similar to RAID 5, but it includes a second parity scheme distributed across the drives in the array.
- The use of additional parity enables the array to continue functioning, even if two disks fail simultaneously.
- However, this extra protection comes at a cost. RAID 6 arrays often have slower write performance than RAID 5 arrays.



RAID Levels

Category	Level	Description	Disks required	Data availability	Large I/O data transfer capacity	Small I/O request rate
Striping	0	Nonredundant	N	Lower than single disk	Very high	Very high for both read and write
Mirroring	1	Mirrored	$2N$, $3N$, etc.	Higher than RAID 2, 3, 4, or 5; lower than RAID 6	Higher than single disk for read; similar to single disk for write	Up to twice that of a single disk for read; similar to single disk for write
Parallel access	2	Redundant via Hamming code	$N + m$	Much higher than single disk; higher than RAID 3, 4, or 5	Highest of all listed alternatives	Approximately twice that of a single disk
	3	Bit-interleaved parity	$N + 1$	Much higher than single disk; comparable to RAID 2, 4, or 5	Highest of all listed alternatives	Approximately twice that of a single disk
Independent access	4	Block-interleaved parity	$N + 1$	Much higher than single disk; comparable to RAID 2, 3, or 5	Similar to RAID 0 for read; significantly lower than single disk for write	Similar to RAID 0 for read; significantly lower than single disk for write
	5	Block-interleaved distributed parity	$N + 1$	Much higher than single disk; comparable to RAID 2, 3, or 4	Similar to RAID 0 for read; lower than single disk for write	Similar to RAID 0 for read; generally lower than single disk for write
	6	Block-interleaved dual distributed parity	$N + 2$	Highest of all listed alternatives	Similar to RAID 0 for read; lower than RAID 5 for write	Similar to RAID 0 for read; significantly lower than RAID 5 for write

All tools Edit Export E-Sign

Floppy Disk